

Land - 10'

☕ Coffee/tea 🍵

Installation check, housekeeping - 15'

- ✅ Roll call + 📜 Code of Conduct
- 🖥️ Did everyone install UNIX Shell?
- 🙋 Getting help (🛑 red ✅ green stickers)

Icebreaker - 5'

A short icebreaker from [resources document](#)

Introducing the Shell - 10'

- 👤 introduce UNIX shell using [slides](#)
- demo how to open a shell
 - 🛑 red ✅ green wait for everyone to open shell
- explain \$ and **text cursor**
- explain username e.g. `nelle@localhost`

File systems - 10'

- **File browser:** demo the actions that will be covered in directory navigation using the file browser.
 - This visual explanation should make it easier when moving to the shell.
- Explain concept of `home`
- Explain meaning of slash at the end `/`

Symbolic link to Desktop - 10'

- Users with Windows 10 will probably save the data in `OneDrive` Desktop.
 - First make a symbolic link to make our lives easier.
 - `ln -s "/c/Users/[username]/OneDrive\ -\ Delft\ University\ of\ Technology\Desktop" "Desktop"`
 - Explain how **how spaces need special handling**

Navigating Files and Directories

Type along - 20'

```
pwd # (Print working directory)
ls -help # (Ask for help about the 'ls' command) (Windows / Git-bash)
man ls # Show the UNIX manual page for the 'ls' command (Mac / Linux)
clear # Clears the terminal screen
ls -l -h # Long-form listing of the files in the current directory
ls -F # for MAC
ls -F . # . means the current directory
```

```
ls -F Desktop
ls -F Desktop/shell-lesson-data
cd Desktop # Change directory to the one called "Desktop"
cd shell-lesson-data
cd exercise-data
pwd
cd shell-lesson-data # Error: no such directory
cd .. # .. means the directory above the current one
pwd
ls -a # list all files, including hidden files
cd ~
```

Challenges **filedir** - 10'

- go to [TuDelft Vevox](#)
- start poll -> wait for answers -> discuss -> next question

Break - 10'

Working with Files and Directories

- Talk about **good names for files and directories**
- Explain: create files with **touch**

Type along - 25'

```
pwd
cd ~/Desktop/shell-lesson-data/exercise-data/writing
ls -F
mkdir thesis
ls -F thesis # Empty directory
mkdir thesis # Directory already exists
cd thesis/
nano draft.txt # Edit the file draft.txt
touch my_file.txt # Create an empty file my_file.txt
ls
cat my_file.txt # Show the contents of a file
rm my_file.txt
ls
mv thesis/draft.txt thesis/quotes.txt # Move / rename a file
ls thesis/
mv thesis/quotes.txt .
ls
ls thesis/
cp quotes.txt thesis/quotations.txt # Copy file to a different location /
name
ls thesis/
ls
cat thesis/quotations.txt
cp -r thesis/ thesis_backup # Copy recursively - will copy folder
structures
```

```
ls thesis_backup/  
rm -r thesis_backup/ # Remove contents and folder
```

Challenges **create** - 10'

- continue with the same vevox
- start poll -> wait for answers -> discuss -> next question

Using wildcards (*,?) to access multiple files

Type along - 10'

```
clear  
cd ..  
cd alkanes  
ls  
ls *.pdb  
ls p*.pdb  
ls ?ethane.pdb  
ls ???ane.pdb  
ls *.pdf # No such file or directory  
clear
```

Challenges **wildcards** - 15'

- continue with the same vevox
- start poll -> wait for answers -> discuss -> next question

Key points - 15'

Feedback - 5'